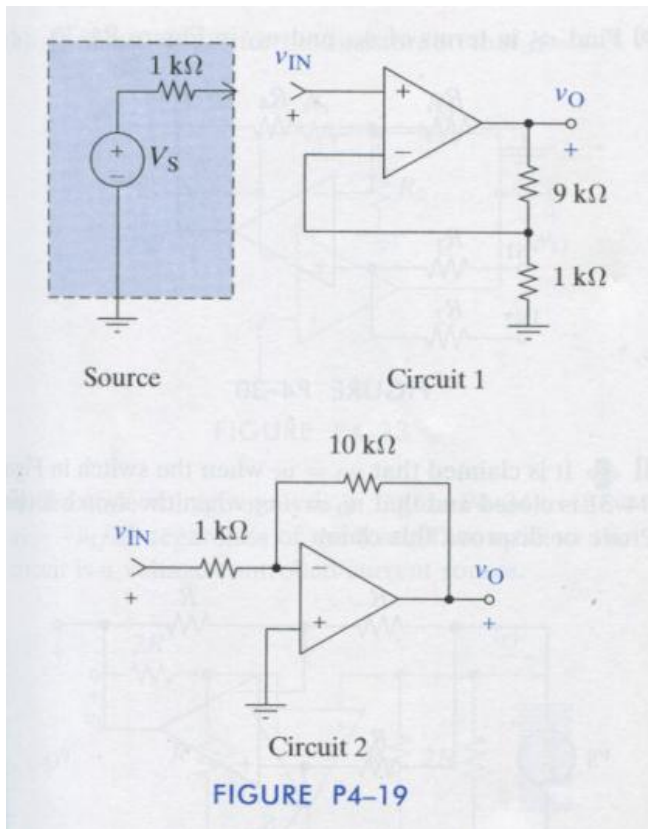


MAE 140 Winter 12 Homework #5

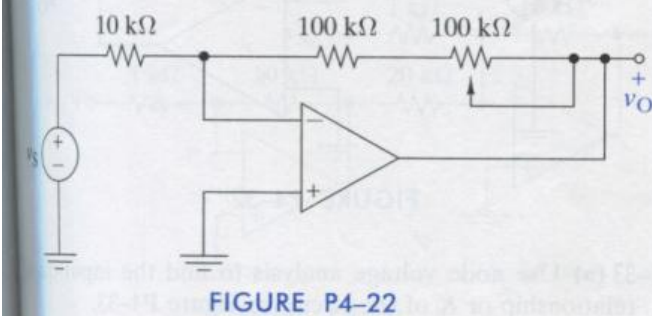
T R & T, 4.20, 4.22, 4.24, 4.26, 4.28, 4.30, 4.36, 4.38, 4.42, 4.44

Due: 02/21/2012

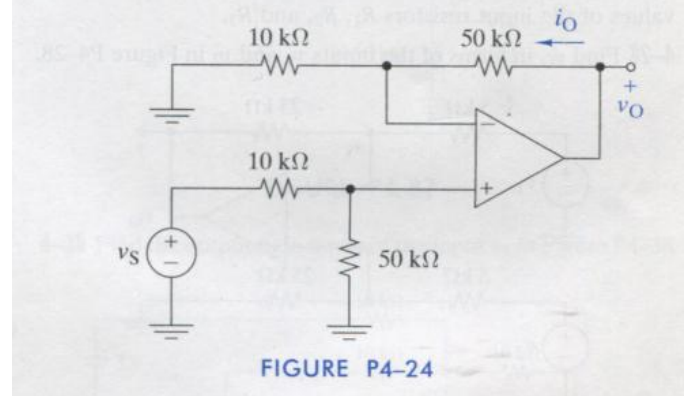


4-20 **E** Suppose the output of the practical source shown in Figure P4-19 needs to be amplified by -100 and you can use only the two circuits shown. How would you connect the circuits to achieve this? Explain.

4-22 What is the range of the gain v_O/v_S in Figure P4-22?

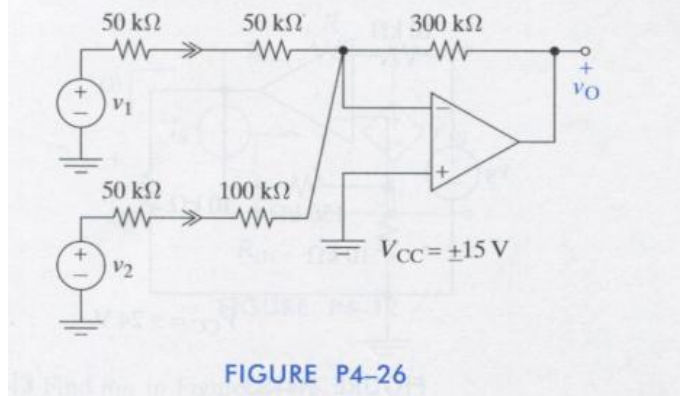


4-24 (a) Find v_O in terms of v_S in Figure P4-24.
(b) Find i_O for $v_S = 2\text{ V}$.



4-26 (a) Find v_O in terms of the inputs v_1 and v_2 in Figure P4-26.

(b) If $v_1 = 1\text{ V}$, what is the range of values v_2 can have without saturating the OP AMP?



4-28 Find v_O in terms of the inputs v_1 and v_2 in Figure P4-28.

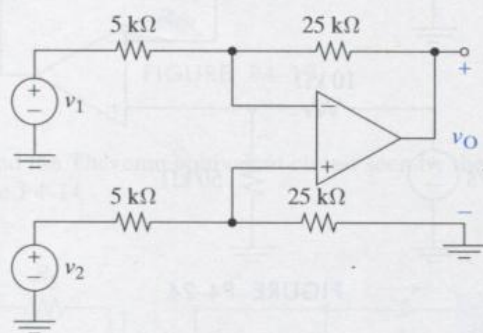


FIGURE P4-28

4-30 Find v_O in terms of v_{S1} and v_{S2} in Figure P4-30.

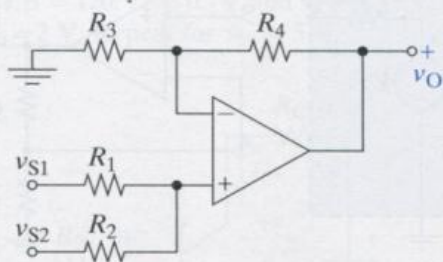


FIGURE P4-30

4-36(a) Find v_O in terms of the inputs v_{S1} and v_{S2} in Figure P4-36.

- (b) Redesign the circuit using only one OP AMP.
(c) Validate your design using OrCAD.

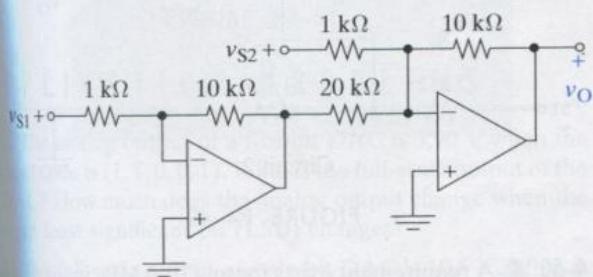


FIGURE P4-36

4-38 Find the output v_2 in terms of the input v_1 in Figure P4-38.

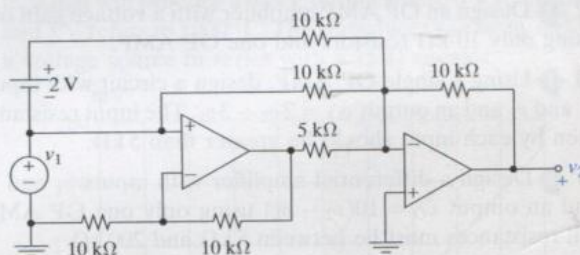


FIGURE P4-38

4-42 Design an OP AMP amplifier with a voltage gain of 4 using only 10-kΩ resistors and one OP AMP.

4-44 Design a differential amplifier with inputs v_1 and v_2 and an output $v_O = 10(v_2 - v_1)$ using only one OP AMP. All resistances must be between 5 kΩ and 200 kΩ.